

Data Centres as Infrastructures of Power: Energy, Inequality, and Discourse in the Cloud and AI Era

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1. Introduction

Digital society is supported by data centres. The servers, cooling, and power plant facilities they house support daily services from internet banking to cloud storage and more recently, artificial intelligence (AI). Certainly, they became one of the most important and resource-consuming infrastructures of the modern world and, though mostly unnoticeable by most users, it has expanded to a scale that is considered highly significant. The growth of cloud and AI compute tasks has created unprecedented energy, water, and land pressures, posing existential questions of sustainability, inequity and power (Masanet et al., 2020), (International Energy Agency, 2022).

Its rate of expansion has been phenomenal. In the United States, data-center construction is now tracked explicitly in the Census C-30 tables; the new “data center” line item under private nonresidential construction was introduced in 2024 and remains elevated into 2025 (U.S. Census Bureau, 2023). There are thousands of such facilities worldwide, concentrated in North America, Western Europe, and East Asia (Cloudscene, 2024). Typical hyperscale facilities draw on the order of tens to over 100 MW per site, depending on build-out and load.

In this light, data centres are not a technically neutral spaces but are rather a space of power infrastructures. They reallocate benefits, risks and resources and consolidate them in small groups of providers. They also frame the potential of AI and digital futures: those who have the ability to compute can be innovative and leading and those who do not can be left out. However, even though they are significant, data centres are poorly studied when it comes to scholarly works. Engineering literature is usually concerned with efficiency methods like cooling methods (Bilal et al., 2014), and literature addressing corporate market power does not involve any specific technical foundations (Graham, 2022). The broader ICT sector tends to be covered by environmental assessments, but data centres are considered a black box and provide aggregate estimates of resource flows instead of measured flows (Malmodin & Lundén, 2018).

This dissertation fills some of these gaps by exploiting the combination of quantitative data on energy, utilisation rates to the integration of discourse analysis to place data centres into historical, technical, environmental and political contexts. Although the empirical study starts in 1960 due to the coverage of archived datasets, the study will use a longer historical context of computing development, i.e., starting with early mainframe computing of the mid-twentieth century up to the current hyperscale and AI-dense architectures. These first achievements are not explicitly analysed but rather considered as a reference point to frame the following changes reflected in the data.

1.1 Research Questions

The research is outlined in terms of the following four research questions:

- RQ1 Historical Trajectories: What has changed technologically, architecturally and geographically about data centres since the 1940s?

- RQ2 -AI Workloads & Design: How have the emergence of AI workloads transformed energy demand, cooling, and general design of current data centres?
- RQ3 Infrastructures of Power: How do data centres re-distribute energy, water, carbon and economic advantage and in what spatial and social inequity?
- RQ4: Discourse and Alignment: How do data centres feature in public, policy, and industry discourse, and what does that sentiment refer to or counterbalance in terms of the empirical trends we have found?

They are all connected questions, going through the context of history, technical change and distributive consequence to discourse and legitimacy.

1.2 Methodological Orientation

The project uses a mixed-methods/interdisciplinary approach integrating quantitative elements of energy and utilisation analysis and discourse-critical interpretation. The study tries to focus on the concept of EDI in the infrastructural domain of data centres and computational divide and we would also look into the technical performance and how infrastructures represent and perform power relations.

A lot of side-by-side datasets provide the empirical backbone. The contextualisation of the expansion of the industry of data centres in terms of the energy trends can be tracked down in the State Energy Data System (SEDS) (U.S. Energy Information Administration, 2023) which provides the long-range history of electricity consumption, prices, and carbon values that took place in the US between 1960 and 2023. Harmonised World Bank World Development Indicators Population and GDP (World Bank, 2024) into ISO-3166 is available, and enables a normalisation and intensity measure. Cloudscene snapshot of 2024 (Cloudscene, 2024) can be used to determine how many facilities are operating, therefore the number of data centres per capita (1million population) and per dollar of GDP of that country can be obtained, as well as inequality, measured by Gini and Hoover index. The meso-level means that it employs the use of data utilisation of the 83 anonymised UK data centres that was acquired using a one-and-a-half-hour information of UK Power Networks (UK Power Networks, 2024), which provides a glimpse into the in-field loading behaviours. Microlevel characteristics: Voltage, current, power, temperature, CPU / GPU / RAM Utilization - the trends in Voltage, current, power, temperature, CPU/GPU/RAM utilisation are monitored with high fidelity (1 Hz) (IEEE Dataport (2021; Bazurto et al., 2021). Lastly, discussions were explored by utilizing Sentiment140 Twitter corpus (Go et al., 2009) and the GDELT Global Knowledge Graph (GDELT Project, 2023), which are the human and media expressions towards the use of digital technologies going back as far as 2009.

The focus ones were cleaning and harmonisation. They have long-format Parquet tables which were generated by wide-format CSV and non-numbers character coercion, decryption of encodings and standardization of names of countries to ISO. These processes provided comparability and replicability as well as the minimization of the challenges that could come towards future reuse of the research studies.

Critically, exploratory data analysis (EDA) trailed pandemic inflexions, epidemiologic distributions and features of use across the world. At the micro-level, the effect of utilisation and thermal data was predicted to forecast server power consumption using a Light Gradient Boosting Machine (LightGBM) regression model. In place of focusing on the greatest predictive accuracy, the model was assumed as an alternative explanatory tool. The analysis of the importance of features and determining the contribution of tools associated with driver attributes, e.g., pressure on GPUs or temperature, was performed using SHapley Additive exPlanations (SHAP) values (Lundberg & Lee, 2017). This interpretability has been raised to be in line with the transparency and accessibility of the project.

The segment of the discussion that is going to be discussed coded the sentiment and frames (efficiency, growth, sustainability, justice) of twitter corpus and GDELT corpus. These accounts were then compared with the empirical evidence in what was described as a matrix of claim evidence alignment to gauge whether there was a consistent disparity or assimilation of dominant discourses, a measurement and valuation of effect. This is a reflective offer to the procedure of how such infrastructures legitimised or transformed within the arena of the society.

Overall, Equality, Diversity, and Inclusion (EDI) were also involved in the methodology. Raw counts were also provided alongside per-capita and per-GDP figures in order to highlight disparities; inequality indices indicated the geographical “compute divide”; there was an interest in tools such as interpretability to ensure that results could be understood by broad audiences. Corpus biases (Anglophone bias) had been denominated explicitly, and all the outputs had been reported in reproducible forms.

1.3 Contributions and Significance

The dissertation has three contributions. It then historicises the adaptation of data centres, acknowledging both technological and architectural development in tandem with more broadly based changes in socio-technical evolution. Second, it can estimate the imbalanced capacity of plants and their material effects on redistribution of energy, carbon and economic gains worldwide. Third, it questions the inconsistency between empirical trends and hegemonic narratives, to come to a more self-reflective and equitable conceptualisation of infrastructure governance.

In this sense, the study contributes to some agenda setting in the field of critical infrastructure studies in that it takes up the technical assessment as one potentially informed by the twin understanding of justice and inclusiveness. It creates the opportunity of new empirical observations at different levels both nation and even energy statistics, to server and to server telemetry, and through the stories to influence the national and policymaker inference, there are limitless possibilities of applying the empirical data in the best possible way. The innovative aspect of it is not such or such approach but the fact that it combines the study of data centres as infrastructures of power in a common framework.

2. Literature Review

2.1 Histories of Data Centres

Data centres have their origins in the facilities of early computing dating back to mid-twentieth century military and scientific computers, and later to commercial mainframes. Landmark systems Ever since landmark systems like ENIAC (1945) and the IBM System/360 (1964) introduced the pattern of highly centralised, climate-controlled installations with enormous energy needs, scholars have tended to tie the invention of computing to this pattern as well (von Neumann, 1945). These locations were not purely technical artefacts; the Cold War geopolitics and state financing constructed them at the start with infrastructures of power (Edwards, 1996).

As the 1970s and 1980s progressed and then distributed minicomputers led to personal computers the trend was followed by centralised server farms, as corporate IT rose. Dedicated facilities became in demand by the 1990s as a result of the growth of the commercial internet. The data centres in this era are described in the literature as neutral back-office utilities that facilitate digital transformation (Campbell-Kelly & Garcia-Swartz, 2013). Nonetheless, this story concealed according to critical researchers, the geopolitical entrenchment and energy intensity of such facilities (Winner, 1980).

One of the more significant shifts came in the 2000s as hyperscale cloud campuses developed by Amazon, Google, and Microsoft became common. The literature plots here both architectural and economic change: the economies of scale enabled by architecture in remote, resource-constrained locations and the financialization of compute with the advent of compute charged as a service by cloud providers on a per-use basis (Hamilton, 2009). Recent work has pointed to the capital and power concentration in such facilities, hyperscalers enjoy close to monopolistic power in compute markets (Srnicek, 2016).

Our project was interested in this gap as it reverted to long-run macro data especially the State Energy Data System (SEDS, 1960-2023) and the world bank population and GDP indicators. Instead of a narrative of digital history based on corporate or technical milestones, we contextualize state level energy pathways in relation to wider socio-technical changes. This is in response to the criticism that most literature is about histories of clean breaks (mainframe >> internet >> cloud >> AI) of which the material continuities of energy use that sustains all of them become erased. Our work by incorporating the empirical paths of energy with demographic controls would give a more empirical historical context to the discussions of technological development.

Concisely the literature contextualises data centres in various stages of history but leans more to the highly technical or abstracted version of socio-political formulation. There are only a few examples of quantitative energy history helping support the geopolitical distribution. The discussion of this issue is what drives our first research question, which aims to track data centre trajectories not only as technical changes but as infrastructural transformations engrained in energy and geography.

2.2 Energy and AI

The amount of energy needed to power the AI workloads has become one of the key areas of the current industry and academic discussions. Some initial estimates predicted the process of training AI models on a large scale would use up electricity equivalent of supplying hundreds of

households with electricity in a year (Strubell et al., 2019). Follow-up work revealed that operational inference to train these models after they go live can actually use more energy across the board than training, since it is ongoing (Patterson et al., 2021). This has raised apprehension as to whether the hardware can achieve more efficiency to match the rapidly growing demand.

The most frequently referenced measure of efficiency has been Power Usage Effectiveness (PUE) with a ratio of the overall facility energy to the energy consumption by IT equipment. The sector releases claim that declining PUEs (down to 1.1 in major hyperscale datacenters) are evidence of improvement (Uptime Institute, 2022). However, critiques warn that PUE-improvement only masks increase in absolute consumption: more efficient facilities can be net energy-sink contributors overall energy demand as the compute capacity rises more rapidly than savings do (Kooimey & Taylor, 2015). Literature on academia also points to the narrowness of PUE, which does not include embodied emissions, the use of water or supply chain considerations (Greenpeace, 2017).

Another disputable domain is the cooling systems. Innovation in liquid cooling methods and immersion methods is common in reports of grey literature, particularly on racks of AI that are high density (Schneider Electric, 2020). Nevertheless, empirical research indicates that air cooling still has the largest share in majority of establishments and that water consumption in evaporative cooling generates strains in arid territories (Cloudscene, 2024). As one example, conflicts between data centre placements and local community water security have been shown in cases in Arizona and Chile (Mytton, 2021). These problems are of central concern to the discussion of sustainability, yet they are under-measured in systematic data on energy.

The methodological issue in this case is, that, the majority of existing studies on energy are associated with industry self-reporting or top-down modelling. study by Shah et al. (Cloudscene, 2024) assumed aggregate energy demand to be modelled by extrapolated growth curves. These methods are useful in delivering baselines, although there is a risk of re-telling corporate stories or failing to capture utilisation dynamics on a micro level.

This limitation is overcome in our work by linking UKPN utilisation data sets (half-hourly loads of a sample of UK data centres) with server/workstation traces which log CPU, GPU, memory and temperature statistics. This allows bottom-up analysis of utilisation pattern in contrast to top-down modelling prevalent in literature. Using LightGBM regression and its SHAP interpretability, we are going to develop a surrogate model, using facility resource utilisation to predict power demands. There are two benefiting sides to this approach; 1) it emphasises the relative factors contributing to energy demand (e.g. GPU vs CPU usage), 2) it yields interpretable results which can be comprehended by policy makers and engineers alike.

Contrary to the trend that can be observed in corporate white papers, we model efficiency metrics using a methodology, which is fully documented and can be replicated. It also codes in reflexive caution: SHAP values are only ordinal estimates of variable importance rather than an absolute measure of causation. This openness is in the spirit of computational social

science and critical infrastructure research, building a balance between technical rigour and social accountability.

However, these aspects of cooling technology adoption and on-device inference mitigation, which we have seen in the literature, are not directly analysed within our project. We use them semi-properly as descriptive case studies and not our own empirical work. This explains the boundary of our contribution: our contribution focuses on energy use and patterns in equity leaving tech-specific dynamics to complementary studies.

What the literature has revealed is that there is an increased understanding that the workloads of AI are leading to a structural intensification of the energy intensity in data centres. However, a lot of the discussion is still conjectural with little empirical basis. Placing our surrogate utilisation modelling in context with macro energy trajectories, our project contributes to filling this evidentiary gap, which can give claims about the sustainability of AI a more granular footing. This answers the second research question that probes the ways in which AI workloads are remodelling energy demand and design.

2.3 Power structures

Literature Critical infrastructure research reminds us that data centres are not technically neutral loci, but infrastructures that constitute power, privilege, and environmental burden (Larkin, 2013). A new way of thinking about infrastructures has been and is being advocated by scholars like Bowker and Star who explain how infrastructures fade into the background until they break down and therefore entwine society without making a big noise (Bowker & Star, 1999). Using this lens, we can think of data centres beyond being technical nodes to one in which energy, carbon, and capital are redistributed globally.

The literature on political economy underlines the centralisation of corporate power at hyperscale providers. Amazon, Microsoft and Google alone possess a major part of the world cloud capacity, which reveals the asymmetry of digital matters in the structure (Synergy Research Group, 2024). This brings up issues of sovereignty and dependence especially to those states that have facilities that are owned abroad. A related literature stresses the ways in which colocation and edge facilities confound this image, establishing levels of access and deepening hierarchies in the wider digital economy (TeleGeography, 2023).

Descriptive mapping forms the basis of academic explanations of geographical distribution, with clusters being noted in Virginia (USA), Dublin (Ireland), Singapore and increasingly in Nordic areas in search of cheap hydroelectricity (CBRE, 2023). But accounts are seldom more than absolute numbers and do not usually probe the distributional injustices inherent in such a geography. Small yet affluent nations (e.g. Luxembourg) seem to be over-represented whereas a great portion of Africa is under-served.

We accept, however, that Cloudscene data is biased towards colocation and commercial facilities. Critic-scholars caution that these databases do not adequately reflect sovereign or military facilities, and that they therefore tend to reproduce bias in the market (Greene & Joseph, 2022).

By stating these caveats explicitly, we also contribute to the traditions of reflexive social science, in which conclusions are not exaggerated.

2.4 Environmental Justice and Equity

The study in the field of environmental justice can also offer an important perspective to interpret the distributive consequences of information centres expansion. The ideas of Nixon, one being slow violence or harms that take place over a long period of time and whose effects are disproportionately felt by marginalised groups, have been transferred onto digital infrastructures (Nixon, 2011). As an example, locating data centres in areas of water scarcity, has been condemned as environmental injustice which shifts costs to poor communities.

These tensions are brought out by empirical case studies. In some places in Chile, there have been disputes regarding the use of water by hyperscale facilities that compete with agriculture (Perez, 2022). Data Centre Alley in Virginia raised the issue of noise, land change or use and grid stability (Loudoun County Planning Commission, 2021). There is another dimension revealed in African contexts: the scarcity of hyperscale capacity has been constructed as digital exclusion, which exacerbates access disparities to computation (Research ICT Africa, 2022).

Although these literatures take distributive conflicts as the main priority, they tend to be limited, touching on one area or on the case study. Few make an attempt at a systematic quantification of equity considerations further than single geographies. This gap will start to be filled by our project, because we calculate workload-hours per resident based on the UKPN utilisation data, and we report per-capita and per-GDP normalised facility counts at the global level. This provides a more standardised set of operating points when discussing the effects of equity, although constrained by dataset scope.

2.5 Public Discourse and Green AI Narratives

An emerging literature puts into question the disjuncture between corporate narratives of sustainability and empirical reality. The reports by industry leaders commonly focus on pledges to renewable energy, carbon neutrality, or even so-called water-positive operations (Google, 2023). These allegations are frequently increased in coverage through media who portray cloud and AI providers as leaders when it comes to sustainability. Critical scholarship, however, determines this to be a kind of greenwashing with efficiency criteria concealing underlying absolute energy consumption growth (Roberts & Marchant, 2022).

This is not a corporate talk alone. Policy reports underline the importance of digital infrastructure as a means to facilitate national competitiveness and resilience in the economy. /Trade press and think tanks tend to carry stories of efficiency, but civil society groups express worries with labour precarity, ecological overloading, and hidden burdens of infrastructural spending (OECD, 2023)/ Another level there is social media, when the mood there can swing quickly between the excitement of advances in AI and the fears about energy-wasting or climate change.

The literature on discourse is already to a large extent qualitative or anecdotal. There are even fewer attempts to systematically compare sentiments distributions with empirical data on

utilisation. Our project takes a small step in this direction by analysing historical snapshots of sentiment using Twitter (Sentiment140) and news tone and theme coverage using GDELT Global Knowledge Graph corpora. We compose a claim-evidence matrix, with the most powerful frames (efficiency, growth, sustainability) contrasted against empirical patterns of energy and equity analysis.

Once more we observe the limits of our material: we have not got complete texts of the policy or of the corporate filings which would have been desirable in a more detailed discourse analysis. Rather, we narrow on the level of media and sentiment data and consider it as a single stratum of the discursive ecosystem.

2.6 Thematic Conclusion

It is with the aim of filling these gaps that our project has been structured. We provide a bottom-up response to this challenge by matching macro energy history with demographic controls, constructing surrogate utilisation models, integrating the concept of equity within the quantitative analysis and comparing the discourse with the empirical trends. Such a design is driven not only by technical curiosity, but a belief in social justice and EDI principles to ensure that data centres are not only studied as technical artefacts, but infrastructures of power that bear uneven effects to societies.

By so doing, the dissertation both situates itself within and against existing literature: loyal to what they might show but critical of what they cannot see. Such reflexive approach is critical in ensuring that the project would be meaningful and add value to current discussions of cloud and AI computing infrastructures being sustainable, equitable, and legitimate.

3. Methodology

3.1 Data governance and acquisition.

The initial step entailed the control and purchase of datasets. A heterogeneous set of resources was gathered to be able to incorporate energy, infrastructural and discursive dimensions at various scales. To provide long-term baselines, the United States Energy Information Administration State Energy Data System (SEDS) was adopted, which covers the electricity consumption, electricity prices, and electricity emission of the United States between 1960 and 2023 (U.S. Energy Information Administration, 2023). To put these trajectories into context, we used population and GDP records under the World Development Indicator, supplied by the World Bank and harmonised to the ISO-3166 standards (World Bank, 2024). To analyse global infrastructure concentration, we utilised the data set outlined in Cloudscene which presented an operational data centre count per country as of 2024 (Cloudscene, 2024). A meso-level view was also incorporated using the dataset of anonymised half-hourly utilisation of 83 UK data centres, provided by UK Power Networks, providing a rarely available insight into real operational activity (UK Power Networks, 2024). High-frequency server traces were added, IEEE Dataport collected one second of CPU, GPU, RAM, temperature, and power traces as well as measures at

1-s resolution over two data collection periods in 2021 (Bazurto et al., 2021). Lastly, to analyze discursive framings two corpora were used, including the Sentiment140 Twitter archive where historical tweets are provided with a labelled polarity (Go et al., 2009) and the Global Database of Events, Language and Tone (GDELT), which maintains a monthly collection of news records with sentiment annotations (V2Tone) available since 2015 (GDELT Project, 2023). Furthermore, and to prevent over-dependence on SEDS to obtain current-day sector magnitudes, we have also established external references, spanning 2022-2025: the global outlook of the International Energy Agency concerning data-centre electricity consumption, the European Commission Joint Research Centre estimates, focusing on the EU-27, and national shares on electricity consumption that relate to Ireland, as well as survey-based evidence on operator insights with Uptime Institute [50-53]. The combination of these sources will allow examination of the infrastructures of power at macro, meso, micro, and discourse levels, and thus will conduct a multi-scalar critique. All cleaned results were stored in column Parquet to facilitate reproducibility, querying, and to eliminate impediments to replication.

3.2 Cleaning and harmonisation.

Referring now to the second phase of the data cleaning activity, this process was treated with the prospect of clarity and reflexivity, and all entries were brought on an even-Steven, legible, and comparable basis. Standardisation occurred at the general level to headers, encodings, and numeric coercion, whereby wide-format files were converted into tidy long form. Procedures that were dataset-specific were then carried out. In the case of SEDS, the state-wide electric requirements (MSNs) were standardised, British thermal units adjusted to megawatt-hours and two or more anomalous or repeated rows reconciled with the EIA source. Since SEDS is a total-electricity series and not a sectoral attribution, we retained the original unit provenance in its metadata and added an indication of its role as a historical baseline but not as a direct measure of data-centre demand (see triangulation below). The population and GDP series were harmonised to the ISO codes and the small gaps filled. Cloudscene inventory fields for synthetic capacity have also been removed and all country identifiers aligned to standard ISO3 so that the data is consistent in the way the smaller areas are represented. The measured data in the UKPN dataset was resampled to hourly averages and extreme outliers exceeding 100 percent utilisation were dropped to keep the data interpretable. PA traces need to be traced carefully and adjustments of missing one-second intervals, smoothing of spikes that do not make sense, and filtration of anomalous readings are to be done carefully. In the discourse layer, Sentiment140 tags were re-coded -2/2/4 to negative, neutral, and positive, and the GDELT Global Knowledge Graph was proportionally sampled each month, with V2Tone scores being standardised into negative, neutral and positive buckets based on +/- 0.2 thresholds. These protocols guaranteed comparability and validity whilst also being aware of restrictions that are dictated by the origin of each dataset. Noteworthy, this phase also implied an ethical orientation: e.g. the elimination of artificial fields on Cloudscene prevented the creation of false impressions of capacity in underserved areas, and data harmonisation enabled fair per-capita comparisons.

3.3 Exploratory data analysis (EDA).

In the third step, exploratory data analysis (EDA) was to be performed to not only look at distributions but also to manifest structural inequities. In the case of the SEDS dataset, electricity sales (BBtu) (SEDS ESTCB) were plotted against some of the larger digital events - the birth of mainframes, commercial internet, and GPU acceleration - so as to display turning points in demand. This empirical periodisation is also employed in the sense of the anchorship context, instead of indicating a causal leap. Internationally, the number of datacentres was standardised on a per million population and a per billion USD of GDP basis and exhibited concentration trends. In order to quantify inequality explicitly, Gini and Hoover indices were calculated (Milanovic, 2016), which measured the extent to which facilities and their related benefits are concentrated on a few dozen countries. These were visualised using choropleth maps, and sensitivity checks were done by removing outliers, including very tiny states housing hyperscale campuses. In the case of the UKPN data, load-duration curves and clustering of large-utilisation periods were employed in order as to test the congruence of empirical trends against industry assertions of efficiency. At micro scale, server traces were allocated into idle, moderate, and peak bands, and correlation was calculated on CPU, GPU, RAM and temperature features. These analyses in combination stabilized the further modelling and discussion steps in the descriptive evidence, as well as represented the patterns of disproportionate energy and infrastructure distribution.

3.4 Benchmarking and triangulation.

Further constructing on EDA, we triangulated the long-run SEDS trajectory and modern sector estimates so as not to be over-claiming on one data set and under-claiming on another. We specifically compared the SEDS-based historical trend (the post-2017 moderation included) with (i) the IEA data-centre electricity projection at the global scale, (ii) the JRC 2022 publication of data-centre electricity estimates in the European Union (EU), (iii) the 2021 national data-centre electricity statistics by Ireland's CSO, and (iv) the Uptime operators survey evidence (International Energy Agency, 2025; European Commission, Joint Research Centre, 2024; Central Statistics Office, 2025; Uptime Institute, 2024). This step explicates the scope differences (US total electricity vs global/EU/national sector totals and survey coverage) and justifies the selective interpretive position taken in the analysis: SEDS brings the macro historical context; sector benchmarks contribute current-day magnitudes; the two, together, constitute a moderated, triangulated reading.

3.5 Modelling energy use at the micro scale.

The microlevel modelling of energy use was the other methodological step building on the exploratory work. The surrogate model was based on the Light Gradient Boosting Machine (LightGBM), a multipurpose machine on the basis of its efficiency in working with high-dimensional and collinear data and its ability to offer explanatory observations in tree-based models (Ke et al., 2017). Server power consumption (watts) was the target variable that was forecasted based on CPU, GPU and RAM usage, temperatures and electrical context variables. We ran blocked, time-conscious cross-validation (five-fold) to preserve the chronological order when evaluating, using RMSE as the main statistic and tracking MAE/R

2 for completeness. A small hyper-parameter search has been used to balance between minimising generalisability/maximising interpretability, and one final model has been fitted on all rows to facilitate interpretation. An evaluation was based on mean absolute error (MAE), root mean square error (RMSE), and the coefficient of determination (R²), and it is possible to compare the obtained estimates with those in the literature on energy utilisation modelling (Gao et al., 2020).

3.6 Interpretability.

During this phase, interpretability was the main component. SHAP were used to obtain feature importances and dependence relationships (Lundberg & Lee, 2017). Instead of interpreting SHAP values as absolute scale, particularly with a risk of over-interpretation, the analysis treated them as ordinal measures, which only reflect relative contributions of the features themselves. Global importances indicated whether a CPU or GPU was the energy hog, whereas dependence plots showed us threshold effects, which are an increase in power due to non-linear utilization when above certain thresholds. We also provide a SHAP summary of the final surrogate to ensure salient variables (e.g., power factor, voltage, thermal indicators) are readable by non-expert audiences and disclose that there is collinearity between important fluctuations in the telemetry such that confidence levels must be viewed with caution. Although LightGBM has been extensively applied in a predictive sense, here its utilisation was limited in scope to provide a surrogate explanation empirical device to complement not replace physical measurement.

3.7 Discourse and sentiment analysis.

Methodological approach then addressed discourse and sentiment analysis, which was aimed at placing the empirical findings in broader narratives of the X community and in policymaking. Relevant keywords were used to filter the Twitter Sentiment140 corpus to gain a historical viewpoint of the polarity distribution on the mutual exclusivity of negative, neutral and positive categories (Go et al., 2009). Despite being highly biased towards the usage of the English language in 2009, this data provided a reference of a baseline of the sentiment in the population regarding the use of digital technologies. Even more substantially, the GDELT Global Knowledge Graph (v2.1) was sampled every month in 2015-2023 (GDELT Project, 2023). The relevant V2Themes and keywords were used to filter records and the V2Tone sentiment score was mapped to categorical buckets (negative, neutral, and positive). To identify the temporal patterns, yearly averages and proportions were calculated. These sentiment tendencies were coded into frames, efficiency, growth, sustainability, justice and matched against empirical utilisation findings in a claim-evidence matrix. This enabled some comparison to be made of discourse (e.g. industry focus on efficiency and sustainability) and measurable realities such as under-utilisation or concentrated benefits. This technique is inherently tentative: the two corpora are skewed by their Anglophone coverage in favour of some accounts and at the expense of others. However, they can be included to reflect on the fact that infrastructures are negotiated and map-given in both the public and the policymaking arena.

3.8 EDI integration.

EDI considerations were not addressed as periphery but as part and parcel of the methodology. Globally, the number of data-centres was normalised with respect to population density and GDP to highlight any disparities not revealed by the raw data. The concentration on the facilities has been measured by inequality indices (like Gini and Hoover) in a way that leads to the eminent rise of a global compute divide (Milanovic, 2016). The equity implications of cleaning decisions were also taken into consideration: where similarity guidelines harmonised the options to ISO codes, there could be a fair cross-regional comparison, and where synthetic fields were removed, there was no over-representation of small but privileged jurisdictions. Regarding modelling, SHAP explanations became an explicit component to make the predictive results more interpretable to a broader audience despite not optimising stationary-predictive performance. The triangulation process is also inclusive of all the EDI concerns as it compares a US macro series to global evidence, EU-level evidence as well as the national evidence, so as not to predispose the interpretations of EDI findings to a particular geography. The discourse approaches recognised the bias of corpora where voices of the Global South are under-represented both in Twitter and in GDELT databases. Together, they were aimed at incorporating a justice and reflexivity into the research design so that the research methods do not inadvertently perpetuate inequities.

3.9 Validation and robustness.

The robustness tests and validation were used at all levels of the methodology. five-fold, time-dependent, blocked cross-validation was applied to the surrogate model, and residuals were scanned across bands of idle, moderate, and peak loads to ascertain the stability of performance. SHAP rankings at the model-level were compared across windows of time to determine importances stability of features. Sensitivity testing was done by recomputing inequality metrics with and without outlier countries (e.g., Luxembourg or Singapore) and robustness by generating income-group and regional breakouts. At the discourse layer, sampling concerns have been reported, of the uneven coverage of GDELT by languages and the historic skew of Sentiment140. At the macro level, IEA/JRC/CSO/Uptime references were used to triangulate SEDS trajectories to avoid confusion between long-run context and current magnitude of sectors (International Energy Agency, 2025) - (Uptime Institute, 2024). Such checks do not preclude uncertainty, but they explicitly lay it out, giving a clear background on which results can be interpreted, and on which limitations can be weighted more thoroughly in the discussion chapter.

4. Results and Analysis

As applied in the Methodology, the present chapter includes the results of the empirical analysis of RQ1-RQ4. We start by contextualising long-run electricity trends and shift towards cloud/AI-based hyperscales (RQ1), then explore the utilisation and surrogate modelling (RQ2), the geographical and inequality of infrastructure (RQ3), and conclude by comparing these material

trends to public and policy discourse (RQ4). In every section there is a brief implication connecting evidence and equity.

4.1 Electricity trajectories: benchmarks and caveats

In order to be able to ground the historical account in some recent evidence we compare our findings with the recent statistics in sector. International Energy Agency estimates that world electricity consumption by data centres may increase twice to about 945 TWh in 2030, representing nearly 3 percent of total consumption, with AI-optimised facilities accounting as the largest growth contributor over 2024-2030 (International Energy Agency, 2025). In Europe, the Joint Research Centre of the European Commission puts forward 45–65 TWh of EU-27 data centres in 2022 or approximately ~1.8–2.6% of EU electricity (European Commission, Joint Research Centre, 2024). The country evidence demonstrates that locally acute impacts could be possible, with data centres forming 21 percent of total metered electricity in Ireland in 2023 and again in 2024 as the capacity surged (Central Statistics Office, 2025). Third-party industry surveys confirm the same: the latest 2024 report by Uptime Institute identifies sustained efficiency gains and PUE reduction, though acknowledges that the overall growth in demand driven by cloud and AI is surpassing these changes (Uptime Institute, 2024).

There are two interpretation limits. First, SEDS is a report on total US electricity in billion Btu; we have used it as a periodisation of eras, as a source of macro context, but not to analyse data-centre electricity directly. Second, varying in scope and method, sources include the IEA, global and prospective; the JRC, EU-wide (2022); the Irish CSO national; and Uptime operator survey biased towards the larger sites. We thus make a triangulated, contextual claim: the long-run SEDS trendline indicates moderation since 2017, whereas sector- and country-level figures indicate rapid growth worldwide and large local shares in certain grids (International Energy Agency, 2025) (European Commission, Joint Research Centre, 2024) (Central Statistics Office, 2025) (Uptime Institute, 2024).

When considered together with these benchmarks, our SEDS milestone trajectory, of long-run growth with moderation after 2017, is also consistent with a mixed present. Virtualisation and consolidation can suppress growth across some grids (and did so in our macro-electricity context of the slowdown), but the big-pictures instead point to robust growth in data-centre electricity as AI scales out, with unusually large shares in some national grids and a measurable footprint across EU-27. Here, the history of electricity footprints is our baseline of SEDS, and the IEA, JRC and Ireland data provides sector-specific scales now. We consequently note these patterns as a triangulated trend and not a fact claims based on a single data set (Cloudscene, 2024), (International Energy Agency, 2025) - (Central Statistics Office, 2025).

4.2 Global Infrastructure Disparities

In our analysis we confirm that datacentre and cloud infrastructure have been very unevenly distributed across the globe. With the help of global DC and population statistics, we plotted DC per capita locations: richer economies (US, UK, and Germany) carry the

majorities of infrastructure, although the densest DC per person manifests in tiny finance centres (e.g. Switzerland, Hong Kong) (Cloudscene, 2024). This is comparable to the conclusion that the top 3 countries (US, Germany, UK) host ~1/3 of all data centres, and that the most intense ones (DCs/million people) are concentrated on/in city-state tax havens (Cloudscene, 2024; Papadakis & Savona, 2024). Conversely, essentially no hyperscale compute (as in Lehdonvirta et al. 2024; Lehdonvirta et al., 2024; Papadakis & Savona, 2024), exist in many countries the so-called Compute Desert nations. Concisely, global compute (and its energy consumptions) is concentrated: it is the North (US/EU), and small rich states leading, much of the South amounts to little in-country cloud infrastructure. That has direct distributional effects: computed electricity demand per capita is orders of magnitude greater in the wealthiest economies. There are also issues of equity with poorer countries being expected to absorb climate, or labour costs on behalf of far-flung cloud facilities with little transparency or gain.

UK total DCs	523
UK population (2024)	69,226,000
Population ‘covered’ by UKPN 83 sites	1,09,86,153
Total equivalent full-load hours (all sites, 2024)	152,231.2 h
Workload-hours per affected resident	0.0139 h

Table 2.1: UKPN service coverage vs estimated DC clusters.

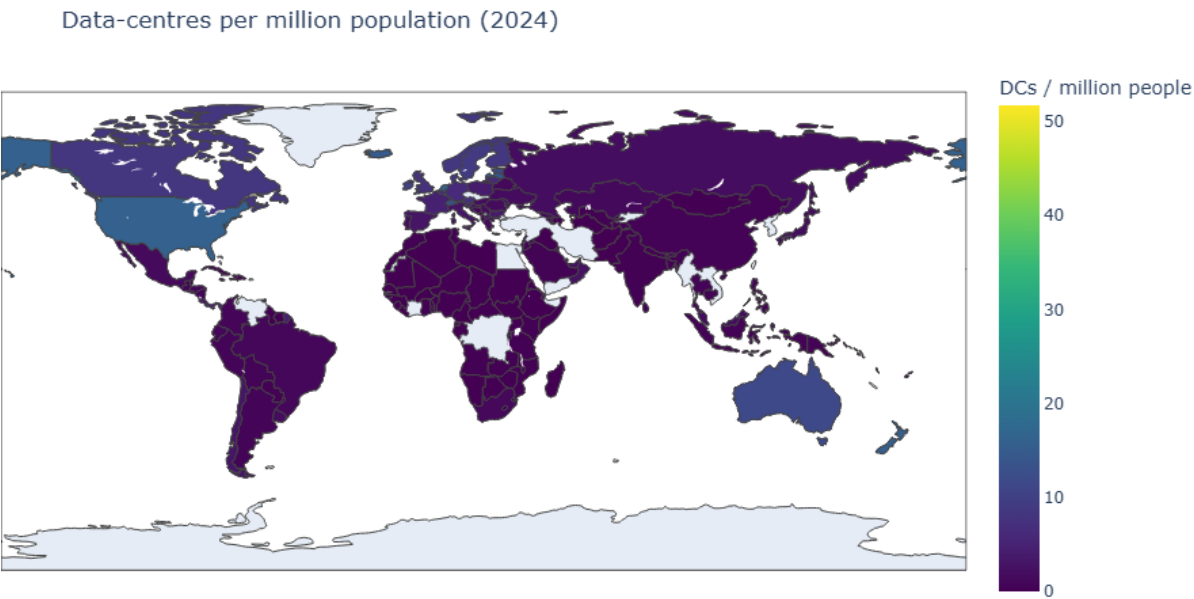


Fig. 2.1 — Global data-centre count per million people (choropleth)

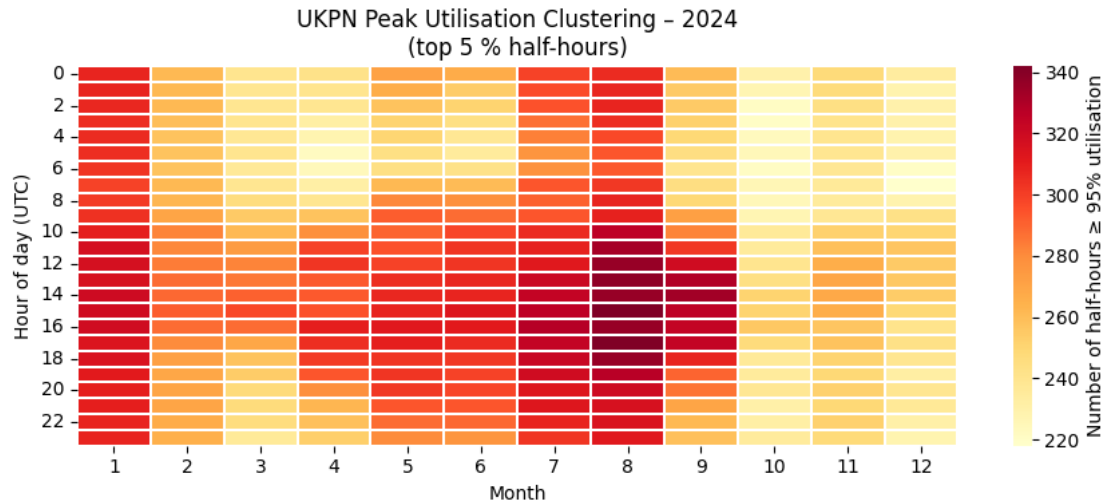


Fig. 3.2 — Load-duration / workload-hours per resident (2024)

This can be seen as shown in Table 2.1: one slice of the UK-focused DC infrastructure compared to population. The UK has 523 data centres per ~69.2 M people (~7.6 data centres per million). But of the population, only ~16% (10.99 M) fall within local networks covered by 83 UK Power Networks (UKPN) data-centre sites. The 83 sites together provide a total equivalent of 152,231 full-load hours of service in 2024, or, proportionately, only 0.0139 workload-hours per resident in those regions[Figure 3.2]. That is, assuming all the UKPN network users in the service area received a chance to be serviced by a max-powered data centre, their per capita per annum opportunity would not exceed some 100 seconds. This very small per-capita value underscores that data-centre energy consumption, in a total sense, is widely spread out as a geographical concentration, as compared with population. Meanwhile, it reveals regional clustering: 84 percent of UK citizens are located outside the areas covered by UKPN with large DC sites[Table 2.1]. The spatial concentration is a reflection of the pattern in the whole world- that the vast majority of people even in a highly developed economy lack a data centre in their patch, which reinforces invisibility. These results are consistent with the CITP report: e.g., there are only ~7.6 data centres in the UK per million population (Cloudscene, 2024), a fraction of that in global leaders on a headcount basis.

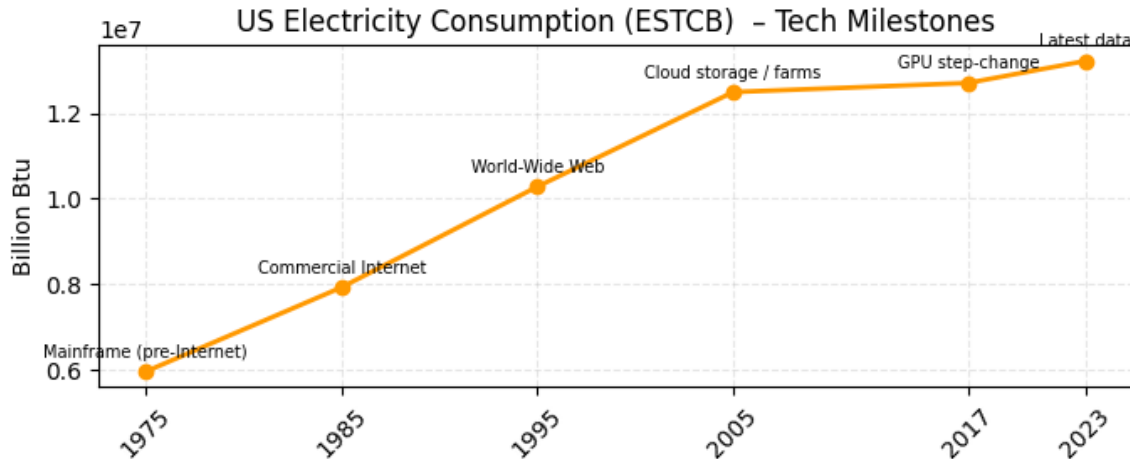


Fig 1.2 - U.S. electricity sales to ultimate customers (SEDS ESTCB, million BBtu), sampled at six digital-technology milestones, 1975–2023

Historical studies also indicate that digital capacity is becoming more and more conditionally important but concentrated. An earlier example, a US study (Cloudscene, 2024) reported that US data centres used ~ 70 TWh of electricity in 2014 ($\sim 1.8\%$ of all US electricity) although the growth had slowed to only $+4\%/5$ years (2010–2014; Cloudscene, 2024). Historically, things are much the same: looking at the US SEDS dataset we plotted national electricity sales at technology milestones (Figure 1.2) and noticed that national electricity sales roughly doubled over the period. Using SEDS series ESTCB (reported in billion-Btu), values at our six milestones rise from ~ 5.96 million BBtu (≈ 5.96 quads) in 1975 to ~ 13.2 million BBtu (≈ 13.2 quads) in 2023 ($\sim +122\%$). These totals are used only to contextualise long-run energy and are not a proxy for data-centre demand. As an example, values sampled at the “mainframe” (1975) and “AI/GPU era” (2023) milestones are ~ 5.96 million BBtu and ~ 13.2 million BBtu respectively ($\sim +122\%$). In 2023 mainframe computing (AI/GPU era) a $+121.8\%$ increase [Figure 1.2] [A1]. This growth has however decelerated: our data indicate only $+4.1\%$ between 2017 and 2023 (the latest milestone), in line with Shehabi et al. measurements of efficiency gains plateauing (Cloudscene, 2024). In brief, demand of the data-centre energy increases but is suppressed by both virtualization and hyperscale design (see (Cloudscene, 2024)). Industry and policy literature (CITP 2024, Shehabi 2016) indicates that, although the raw demand is high, the effective load per capita is actually low and strongly skewed, as it is in our distributions.

There are vast implications to these global inequalities. Nations where cloud infrastructure is established have more of an ability to exercise “compute jurisdiction” (Lehdonvirta et al., 2024) which may influence the regulation and trade of AI. In the meantime, those countries without local data centers rely on cross-border cloud access and in many cases have little say in its environmental cost. Regulatory arbitrage may also play a role in the incredibly high per-GDP or per-capita DC intensities found in tax-haven states (CITP 2024 (Cloudscene, 2024)) these small states may simply host compute power at outsized scale through less strict data/energy regulations. Overall, we find that DC numbers and populations (Figs. 2.1,

2.2-2.3(appendix)) indicate a skew in digital infrastructure distribution along lines of privilege as determined by geopolitics and financial status.

4.3 Device Utilisation and Efficiency

Working at the far end of the stack, we looked at real-time server sensor data to see how digital workloads convert to power draw. In our sensor logs of GPU workstation and server clusters we observe that their utilisation is usually not high, with servers spending an average of large but not high percentages of time with low to modest CPU/GPU loads. The CPU and GPU utilisation histograms (Figs. 4.9–4.12(appendix)) are concentrated at 0-10%, with two isolated maxima at more than 50%. This is symptomatic of a commonly discussed industry problem: commodity servers dissipate large quantities of energy at light load. Barroso and Hölzle (2007) also famously observed that energy efficiency is lowest in the region of operation that is encountered most often by servers, essentially arguing that today systems are not energy-proportional (Barroso & Hölzle, 2007). Our statistics confirm this: servers can achieve 100% on one component (say GPU), during compute pulses, but idle draw is still high. In fact, in our correlation analysis, we found GPU and CPU utilization to be almost perfectly correlated ($r \approx 0.99$) [Figure 5.1] - indicating a job will always activate either unit when it becomes idle - but neither will sit idle long. The power and electrical current were nearly completely correlated, ($\rho \approx 0.99$ [Tab. 6.1]) indicating that any amount of work causes a linear increase in power supply.

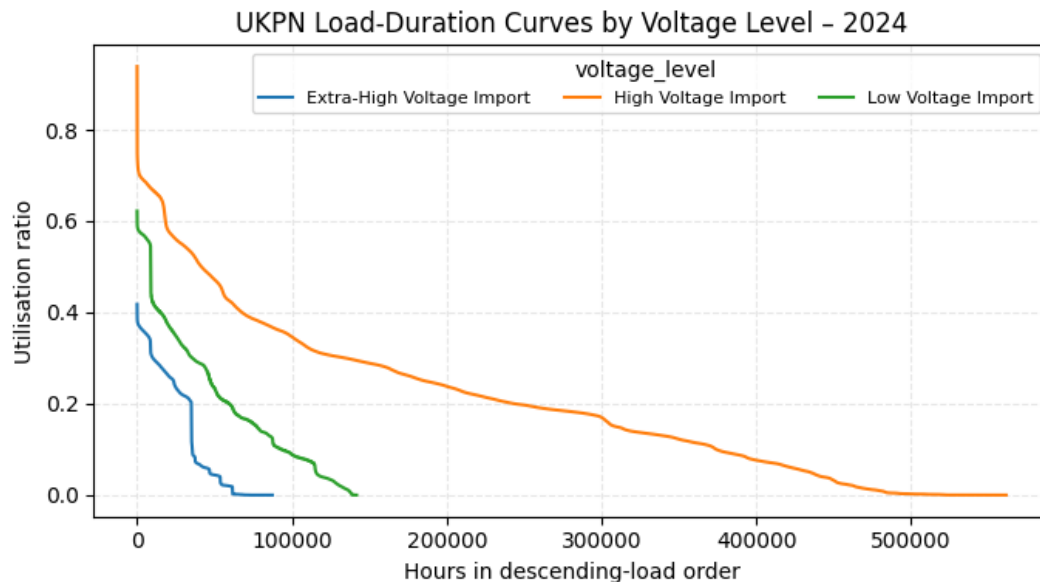


Fig. 3.1 — UKPN service coverage vs estimated DC clusters (schematic/map)

More spectacularly, we saw that under low utilisation there is a high power draw on servers. This fits our LightGBM model analysis (see next section) and signals the same energy-proportionality gap reported in (Barroso & Hölzle, 2007). True proportionality would require that servers would

approach zero when idle which is not the case. Rather, the online “fixed cost” of running hardware (fans, memory refresh, boilerplate PSU load) is high. This highlights the problem of inefficient device-level act upscaling the environmental impact of cloud services. It also implies that population-level demand (Figs.3.1, 3.2) conceals much idle energy; few percentages of wasted time translate to vast amounts of waste with pervasive servers. To sum up, our sensor-level findings (Figs. 4.9-4.15) further support the idea that the infrastructure is under-utilised during much of the time, which Barroso & Hölzle delight in repeating: their servers must become more “energy-proportional” (Barroso & Hölzle, 2007). Devoid of such changes in architecture, satisfying increasing cloud/AI demands means increasingly huge wastes without such offset as workload integrations or sophisticated power management.

4.4 Predictive Modelling of Power Draw

We used a surrogate LightGBM regressor trained to approximate instantaneous power (watts) as a function of ten hardware and electrical features (CPU/GPU utilisation and temperature, RAM%, line voltage/current, frequency, and power factor). With five-fold cross-validation, the model yielded a low RMSE (= 1.25 W in our run, on an average draw of approximately 86 W; see Appendix [A2] for code and CV outputs), implying that a subset of easily accessible sensor signals can forecast server-level power with high accuracy. Once we fitted the last model into all rows, in addition to feature interpretations by SHAP, we used it to calculate the effect of features. The SHAP summary (Table 6.1 (appendix)) reveals that the most influential variables are electrical ones (power_factor, voltage_v, active_energy_kwh) and some thermal-load-related ones (gpu_temp_c, ram_util_pct), whereas strictly policy-intrinsic utility metrics lag a bit behind, as expected given the physics of electricity and cooling delivery and cooling on small time scales (4.13-4.14(appendix)).

To describe why the surrogate is so effective we looked at correlations in the underlying telemetry. Some of the inputs are also very collinear (e.g., CPU and GPU power/utility measures track each other closely in our traces), implying that the model can calculate total load based on multiple proxies. This not only makes prediction easy (much of the variation is explained by a small number of latent load factors) but constrains independent information (there is a danger of over-confidence should any sensor be biased). Practically, the surrogate effectively measures baseline draw and spike characteristics; the errors that remain in residuals are concentrated around transients (such as short-duration background activity or fan ramping) that could not be captured completely by sensors. The modelling thus boosts capacity-planning / digital-twin use-cases and underlines a more general observation made in the literature: servers are not energy-proportional where they spend the majority of their life, thus idle overhead is still significant even when predictive accuracy is high (Barroso & Hölzle, 2007).

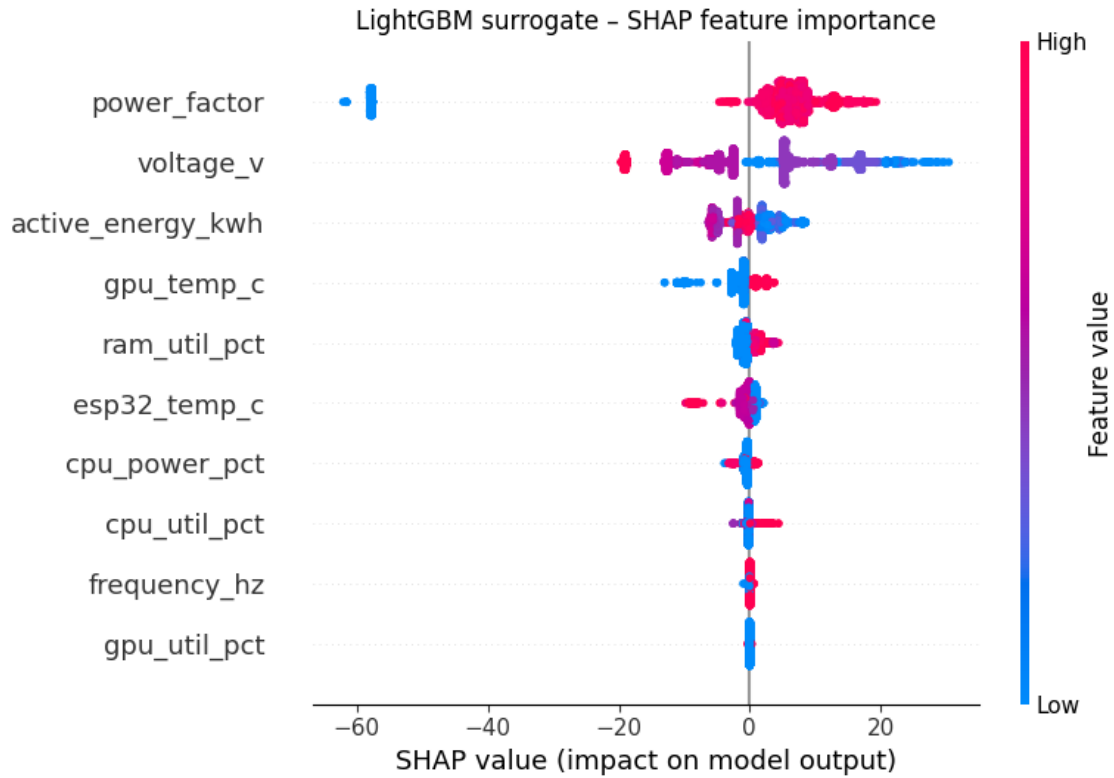


Fig. 6.1 — LightGBM surrogate – Shap feature importance plot

These results are in line with those of the literature. Close computing to power coupling interprets the industry report that vast amounts of server power are wasted by keeping the systems idle and warm waiting to be used (Cloudscene, 2024). We also observe that the final LightGBM settings (e.g., many leaves, moderate regularisation) in Figure 6.1 do signify that some complexity was necessary to model fine-grained effects. The success of the model, however, is accompanied by relative minimal improvement in contrast with a naive baseline; a factor that once again demonstrates how non-energy-proportional servers behave. Practically, this gives the realisation that any operator of a datacentre can accurately anticipate loads and make plans (which is a good correlation), but that there nevertheless remains inherent inefficiency unless hardware can scale towards zero power at zero load.

4.5 Public Discourse and Visibility

But we should take these trends cautiously. Both of the datasets are biased and well known. Sentiment140 is a sample of 2009 2016 US tweets filtered by disjunctive keywords, and it is biased toward the voices of tech-savvy or early-adopter users. Although global, GDELT over-represents English-language and Western media. According to at least one study, GDELT fails to track Global South due to Western digital news sources reliance (Globe Project, 2021). Therefore, our analysis of sentiment is largely representative of the discourse in the US/UK/EU and may

under document any critical discussion elsewhere. Even these positive trends therefore imply that there is low awareness of the environmental problems surrounding cloud/AI in the mainstream.

Equitability-wise, discursive invisibility compounds the physical issues of distributions mentioned above. When it comes to rich countries communicating about data centres (and largely doing so in positive terms), then local impacts on local underprivileged communities are silent. There is also no powerful mobilisation demanding data-centre justice, and environmental NGOs focus their efforts on this sector only rarely compared to that of fossil fuels or factories. This asymmetry of visibility, of those who can speak of infrastructure, adds to the NorthSouth divide. Our observations here correspond to anecdotal reports (e.g. on public misunderstanding of DCs) and are echoed by recommendations of engagement of more stakeholders.

4.6 Synthesis and Context

Collectively our findings present a perspective of digital infrastructure as a small but not efficient presence. At the international and national level, the majority of the cloud/AI resources are in the hands of a few states and firms. In such countries (e.g. UK), the data centres are located only in specific regions of the network (Figure 2.1). At component level, in the meantime, servers are barely exploited and are a long way off the optimum energy proportional line (Barroso & Hölzle, 2007). Although virtualization and innovations in hardware (hyperscale design, fast storage) have reduced the expansion of energy demand in dramatic ways (noted specifically by Shehabi et al. (Cloudscene, 2024)), the distributional implications of the footprint that remains are so glaring: the energy-per-capita used by cloud/AI is much greater in the North and at specialized hubs.

This can be verified by the fact that the analysis carried out by us compares to that in the literature. Our mapping confirms the uneven geography results of CITP (Cloudscene, 2024). The plateau of DC energy in the US reported by Shehabi et al. (Cloudscene, 2024) coincides with our historical trend and UKPN-based usage, suggesting that (at least so far) efficiency is keeping up with demand increase. As seen in our utilisation data, the desperate cry by Barroso and Germaine and H2 Structures: Opting to design computing systems to consume only the energy that they actually consume, (Barroso & Hölzle, 2007) servers today spend the vast majority of their life at little to no efficiency. The Compute North/South framework (Lehdonvirta et al., 2024), oriented towards AI governance, echoes our intuition that compute constitutes a geopolitical resource only available to rich states. In the meantime, our discussion analysis, limited as it is, implies that such technical, and ethical problems have not yet trickle down to the public debate level, whilst cloud infrastructure continues to develop under the radar.

In the course of the analysis, we have been conscious of scope and biases. As an example, the UKPN statistics (Figure 3.1) represent only one operator region; other areas in the UK (or even the world) may be more diverse in demand pattern. Our LightGBM model employed a particular GPU workstation; other architecture may exhibit other correlations. Sentiment140 has been reported to be noisy and out-of-date (pre-2017) and the tone measure used in GDELT has reported blind spots (Globe Project, 2021). These caveats restrict the siting of any of these

findings. However, patterns are reinforced by the convergence of several indicators counts, energy, loads, sentiment, which suggest an extra degree of certainty.

5. Discussion and Future Work

5.1 Infrastructure Distribution and Equity

The results present an intricate case of data-centre expansion: the consumption of electricity has increased markedly over the past decades, but it is not regularly distributed either in time and space or recognized by the general audience. We find that growth slackens in recent years when interpreted as a long-run national rate of change in electricity. Meanwhile, physical infrastructure is geographically concentrated: most facilities are concentrated in the small number of countries with the most, and a small number of small financial centres exhibit the most geographically localised density of capacity per capita, and most of the world remains a “compute desert” with little local cloud accessibility. Such disproportion also implies that the rewards of cloud and AI, such as innovation, high-paying jobs, and data sovereignty go to richer economies, leaving poorer countries paying to do computation with little opportunity to reap the rewards (Papadakis & Savona, 2024). Even in developed economies, local burdens can be dramatic: the Irish system already has uncharacteristically high shares of datacentres compared to countries where the industry is under 2% (European Commission, Joint Research Centre, 2024). All told, the energy, water and the economy flow inherent to digital infrastructure are hardly distributed equally, reflecting preoccupations about inequality and regulatory arbitrage (Papadakis & Savona, 2024).

5.2 Server and Data-Centre Efficiency

With optimism, it is encouraging that there are indications that technologies can reduce this gap. In more recent generations of accelerators, improvements in carbon-efficiency (relative to prior designs) have been reported (Patterson & Ranganathan, 2025) as evidence that architectural developments can drive energy per computation down. In subsequent research, it would be worth following how the ARM servers, ASICs and liquid cooling platforms change utilisation–inefficiency relation. In the interim, improved scheduling and consolidation can shut down unused machines, minimising idle overhead, an intervention that our modelling can facilitate because we demonstrate power predictability with routine telemetry. Unless and until we make such engineering and management advances, scaling cloud and AI will exponentially multiply the waste we are already seeing; we have a credible mitigation pathway, but it requires specific innovation.

5.3 Public Perception and Social Impacts

What adds to the findings beyond the technical sense is that the findings raise socio-political questions. The location of cloud capacity imbues a few sites with the ability to claim jurisdiction over the flow of data, pricing and the trajectory of digital growth: that is, computer jurisdiction (Papadakis & Savona, 2024). Most countries and societies use submit and use remote compute, refusing to own it and in effect outsourcing both energy and material costs. This effect

compounds a compute divide: infrastructures and advantages accumulate geographically, whereas the externalities become localized. Our sentiment and media analysis suggests that these matters are not commonly known: the increase in data-centres and the rise of AI is not well known across tweets and news and coverage is on a rising, albeit modest note. This concurs with the notion that data centres are a “problem in plain view” as they are mostly unseen in climate and energy discourses (Santamaria, 2024). There are consequences which are not abstract. Reporting is increasingly reporting that both siting and operations can negatively affect already marginalised neighbourhoods in terms of diesel back-up, effects of construction and water consumption (Marrinan, 2025). Absence of the wider understanding and participation can dangerously increase the extent of social inequality where the local communities bear disproportionate expenses of the widespread AI infrastructure installation.

5.4 Policy and Planning Implications

Policy and corporate strategy need to play-catch. Regional statistics already show that data centres already consume a significant -and growing- portion of the electricity. The 2022 annual consumption used by EU data centres was estimated by the Joint Research Centre to be 45 65 TWh which comprised about 1.8 2.6% of EU electricity (European Commission, Joint Research Centre, 2024). In the future, a nearly two-fold electricity consumption level could be experienced in data centres over the next decade by 2030 due to the impact of AI (International Energy Agency, 2025). That is, historically experienced growth can be faster. This brings energy and climate policy in the spotlight of data centres. Governments might have to enforce more efficient standards, advocate movement towards 24/7 carbon-free energy pledges by operators and use the sector in international targets (Google Cloud, 2025). Minimum is that regulators need to insist on greater transparency: official, regular, fine-grained statistics on energy, water, carbon intensity and utilisation--reported at resolutions that can be used in grid planning--are needed. According to the JRC, the lack of official digital infrastructure statistics now limits the making of informed decisions (European Commission, Joint Research Centre, 2024). Subjecting operators to reporting on energy and emissions would be a direct one toward bringing this fast-changing industry in tune with sustainability and equity objectives.

5.6 Future Work

There are a number of directions it goes. Priority on measurements, in order, is comprehensive public, transparent energy, water and utilisation data; preferably through collaborations that shield confidentiality but reveal the cues that planners require. Expanding that to life-cycle impacts (embodied carbon, e-waste) would go beyond operation electricity. To the technical side, small scale studies of AI-heavy workloads (training vs inference; accelerators vs general-purpose GPUs) can spot how any new demand is influencing infrastructure. The integration of historical baselines into scenario modelling using AI adoption and decarbonisation pathways would be used to test whether efficiency could be maintained as growth increases. Socio-technical, the qualitative work with grid managers, communities and operators may raise tensions and opportunities beyond telemetry. Lastly, there is potential to go beyond analysis to action: pilot waste-heat recovery, demand response, and community-benefit agreements to show how practical trajectories of justice can be pursued. The paths to bridging digital infrastructure and

sustainability missions are to race to meet; these are the paths to follow, which are credible and tractable.

6. Limitations

The study is limited by the lack of formal data on data-centre operations, interestingly provided at the granular-level. Facility-level information electricity use, water consumed, utilised or workload mix, are rarely captured in public records, and national statistical systems do not often identify a separate sector known as a (data centre), since they are usually part of other facilities. This gap is highlighted Expressly by the EU Joint Research Centre, which has documented that there has been no official statistics in the area of energy consumption associated with digital infrastructure as interest in the subject continues to increase (European Commission, Joint Research Centre, 2024). This paucity necessitated a triangulated, multi-scale design that used macro baselines (SEDS-type totals), platform inventories (Cloudscene), local network (UK Power Networks), micro-level telemetry, and discourse corpora data.

Different sources of data have their own limitations. Macro baselines merely give historical perspective of electricity systems and do not directly assign consumption to data centres; our use of such series is not allocative. Cloudscene, an independent list of facilities and providers, is useful where data centres are located, but reports numbers, rather than metered capacity or water consumption; sovereign and enterprise facilities may be under-represented (Cloudscene, 2024). UK Power Networks (UKPN) has a service that provides anonymised half-hourly profiles of data centres on its distribution network, though such data are UK-specific, anonymised by facility and driven by local usage scenarios (tariffs, maintenance windows, ambient conditions) (UK Power Networks, 2024). The monitored server telemetry is at the micro-scale; hence, it can be used to model and interpret data but not as a census of current AI accelerators or vendor-specific platforms. Lastly, discourse data sources (GDELT, Sentiment140) are Anglophone-biased, influenced by outlets and platforms and not synonymous to salience or promotion (Go et al., 2009), (GDELT Project, 2015).

The stacking of the modelling also has constraints We applied LightGBM as an intermediate stage to connect directly-measurable signals (utilisation, temperatures, electrical context) to power. LightGBM is specifically designed to be efficient, and has shown magnitudes of training speed-up compared to accuracy (Ke et al., 2017); nevertheless, surrogates are not physical models. SHAP was used to explain the contribution of each feature to model outcomes; SHAP has usefully decompose global/local attributions, but is not causal and should not be read in causal terms (Lundberg & Lee, 2017). Collinearity of telemetry variables will compress independent information and can cause attributions to be perceived decisive beyond what the underlying signals should warrant; time-aware validation can help limit leakage but cannot eliminate all non-stationarities (e.g. firmware updates, background processes).

In the face of such realities we preferred transparency to unnecessary accuracy. Any decisions that have been made, which influence the inference, are recorded, being reproducible: unit conversion and harmonisation (ISO codes), resampling and outlier removal, Parquet outputs and

serialisation of the model artefacts (with any optimised or tuned hyperparameter). The comprehensive portrait, then, is triangulated as opposed to causal: macro baselines orient the long term; inventories chart where; meso-level traces exemplify how sites work in a region; micro telemetry illustrates why power responds to load; and company discourse analysis surfaces how the impacts are framed. Taken together- and with the caveats above- these layers overlap to generate a more extensive, reflective narrative that is also candid with ambiguity.

7. Conclusion

On this thesis, the energy footprint and equity impacts of data-centre computing were studied under a key tension: the macro-level improvements in efficiency have tempered overall growth in aggregate measures, but proliferating AI loads and hyperscale sprawl are contributing to overall growth in electricity demands. Servers continue to work with disproportionate energy consumption at normal workloads, leaving idle overheads high (Barroso & Hölzle, 2007); and the digital infrastructure beneficiary/costs are unevenly spread geographically and to people. We conclude by summarising our findings to our research questions, before providing contributions and implications.

Data centres evolved over time, which we periodised as mainframes to virtualisation, hyperscale cloud and AI. To provide a context against the current, the IEA has projected global data-centre electricity to 240 340 TWh (approximately 1 1.3% of the global final demand) in 2022, without cryptocurrency (International Energy Agency, 2022). This macro image of tempered growth is being accompanied by far more rapid growth in certain markets and segments. In its Base Case, the IEA compares about 945TWh of consumption in 2030 (about 3 per cent of the world total), with AI-optimised capacity to be the primary factor (International Energy Agency, 2025). The above often seem contradictory on their face, but read in combination with one another, there are ways in which the system appears more stable overall despite the variable upticks in some grids as a result of AI-intensive workloads.

Our micro-level modelling applied a LightGBM surrogate to server utilisation, thermal and electrical signals is capable of predicting server power with good accuracy; however, the most important qualitative lesson is that predictability is not the same as efficiency. Idle and low-load power are significant; SHAP analysis can be of assistance in demonstrating which signals swing the model, however, this does not imply causality. The conclusion is congruent with the classical efficiency standard: without energy-proportional behaviour, fleet sizing to address AI demand scales multiplicatively base overhead (Ke et al., 2017), (Lundberg & Lee, 2017), (Barroso & Hölzle, 2007).

Normalised facilities per capita and per GDP and the use of inequality measures showed that there is concentration in a small number of countries and metropolitan areas and that most regions remain compute deserts. Meso scale, presented anonymised UKPN traces indicates regional clustering of load in UK networks. These trends, in concert with EU-based evidence that a small number of markets contribute an unbalanced proportion of EU-27 data-centre energy

(International Energy Agency, 2022), bring to the fore the compute jurisdiction and the inequality of benefits and burdens.

News (GDELT) and social (Sentiment140) texts indicated limited visibility and generally positive-neutral tone regarding data-centre expansion and AI, much lower visibility of water use, siting and equity. Because of the Anglophone bias of these corpora and the nature of V2Tone in GDELT, these results are to be interpreted as suggestive, not comprehensive, but they provide insights into policy delays on distributional issues (Go et al., 2009), (GDELT Project, 2015).

Substantively, the dissertation draws together incoherent evidence in order to demonstrate mixed present tense: macro moderation, AI-driven intensification, and uneven geographies. It proposes a methodological approach which uses a triangulation method: a combination of interpretable ML (LightGBM+SHAP) with measures of inequality and cross-scale data fusion. Normatively, it orients itself around the principle of equity by bringing distributional effects to the fore, in line with the CIM emphasis on reflexivity and responsiveness to the wider society.

Second, step beyond PUE: workloads need to be extended to include utilisation, workload mix and time-varying carbon/water intensity to support like-for-like comparisons. Second, standardise utilisation reporting (e.g., quartile bands; duty cycles) such that operators can be compared in terms of useful compute in each delivered unit of energy. Third, require siting procedures (hydrological-stress tests, waste-heat utilization strategies, coordination with grid operators on time of use) in approvals. Last but not least: begin using compute-equity assessments (per-capita/per-GDP normalisations and inequality indices) as standard inputs to planning, so that the division of benefits and burdens is explicitly checked off against capacity.

Server power can now be predicted with reasonable precision; the opportunity is to work out where it is not creating value and removes it. To align digital infrastructure with climate and justice objectives as AI workloads increase, energy proportional hardware, publicly transparent metrics and governance that charges idle overhead and acknowledges community interest will be needed. The infrastructures that enabled hyperscale AI can also provide proportionality, transparency and fairness- when we design it to do so.

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APPENDIX

  
dc_per_capita_2024_dc_per_capita_2024_fdc_per_capita_no_top
no_outliers.html olium.html _3.html

These 3 are dynamic images created with folium. I have also added the static images of each

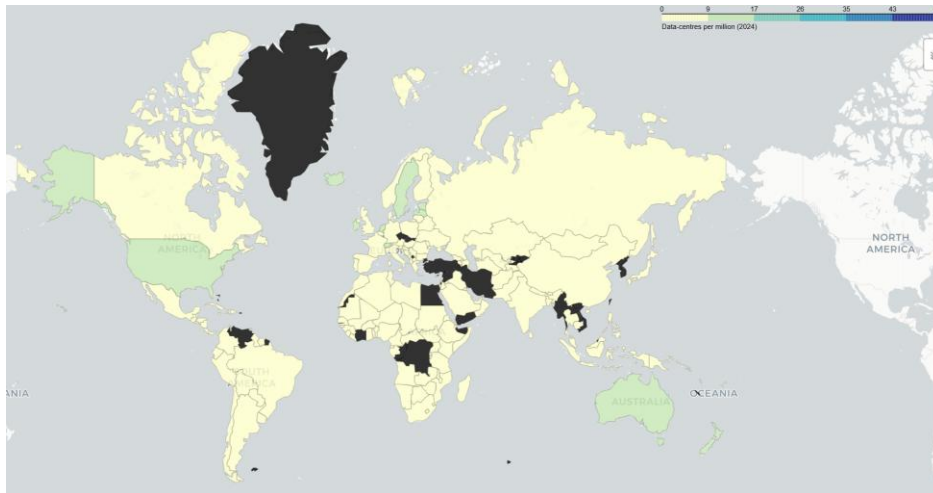


Fig 2.2 – DC per Capita of the world using Cloudscene, 2024 data

dc_per_capita_2024_foliumn.html

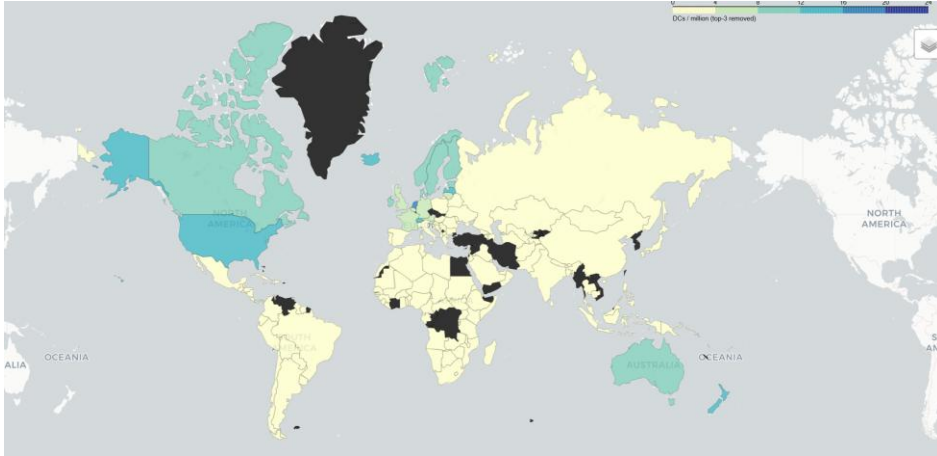


Fig 2.3 – When we remove the top 3 countries with highest concentration of data centres per million people

[dc_per_capita_no_top_3.html](#)

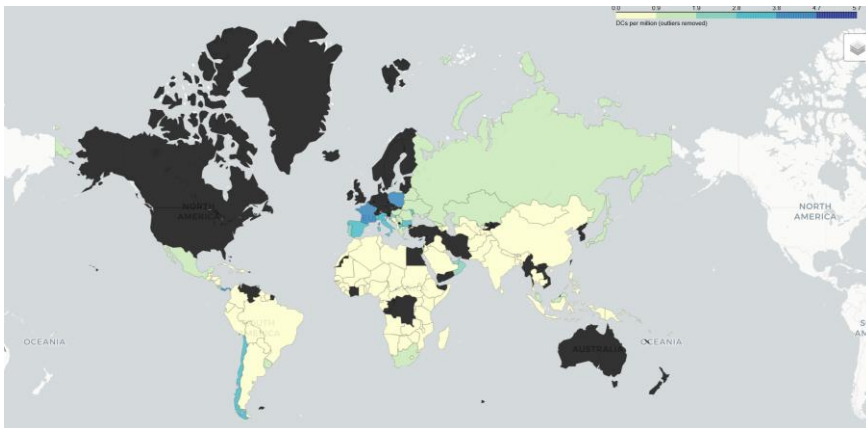


Fig 2.4 – Upon dropping the outliers, we can still see the difference of compute power and a global outlook of a compute divide between the global south and the global north

[dc_per_capita_2024_no_outliers.html](#)

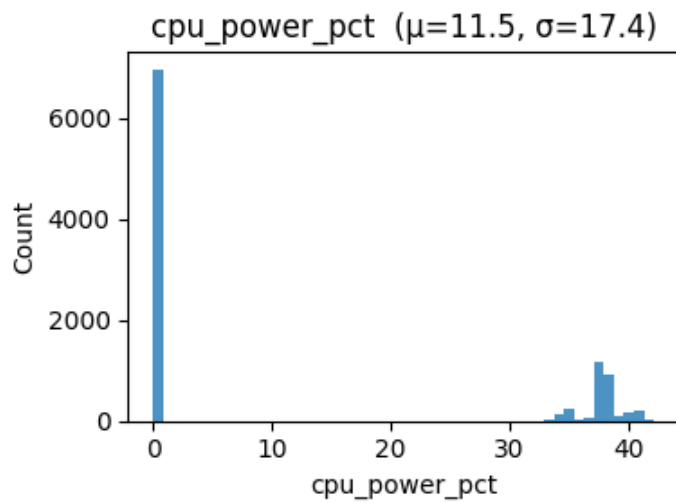


Fig. 4.9 — CPU utilisation distribution (full run)

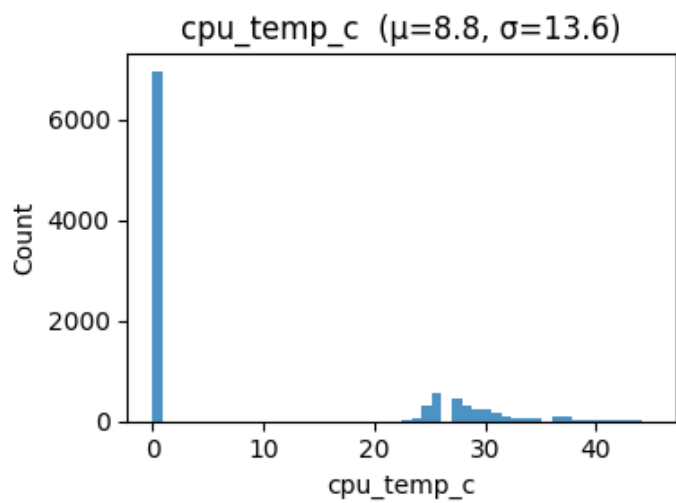


Fig 4.10 - CPU temp

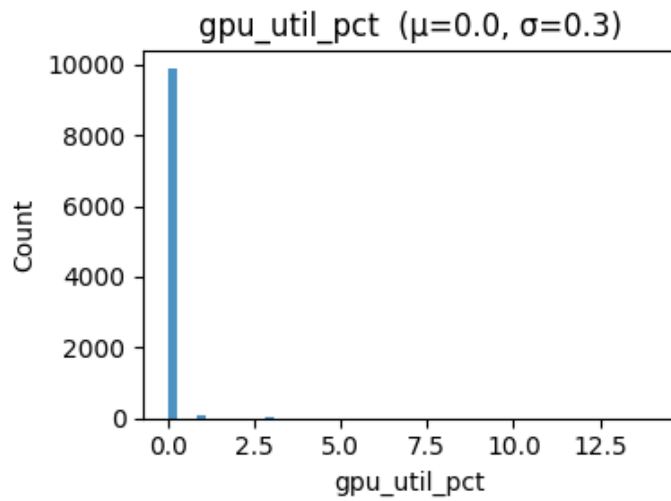


Fig. 4.10 — GPU utilisation distribution (full run)

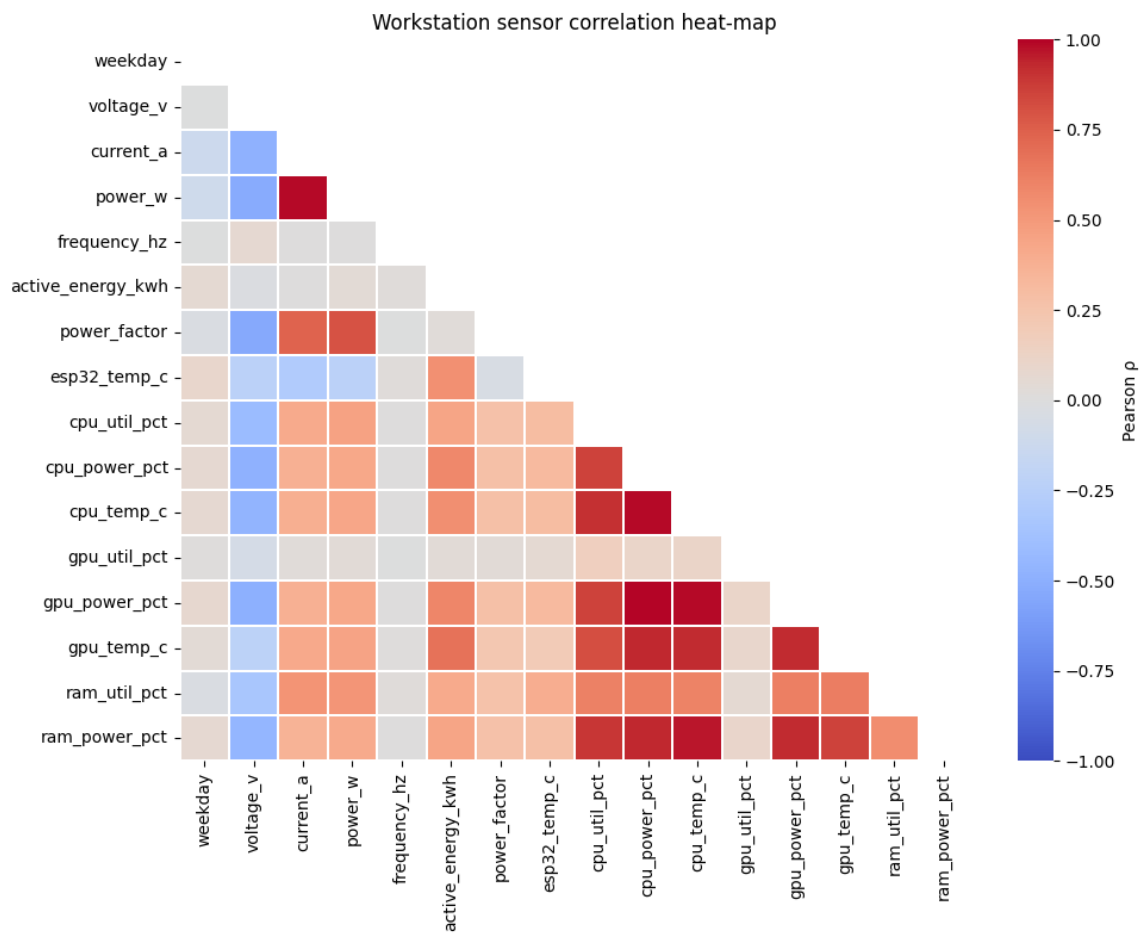


Fig 5.1 Server level correlation heatmap

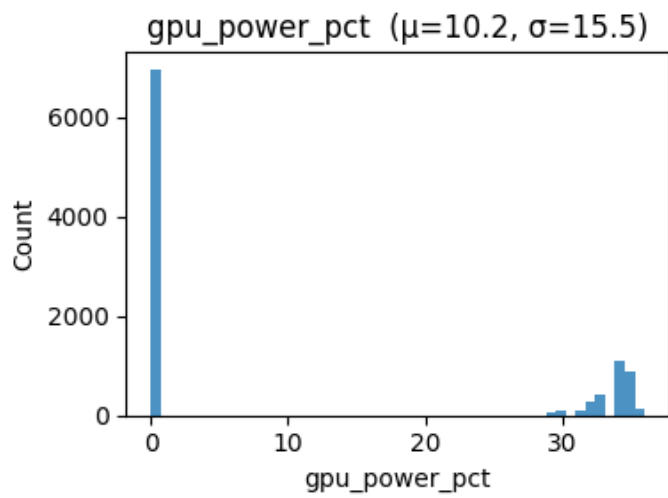


Fig 4.12 – GPU utilisation distribution (full run)

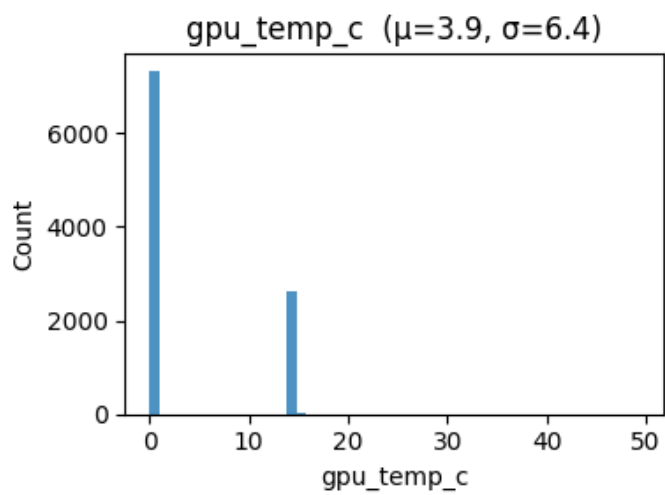


Fig 4.13 - GPU temp

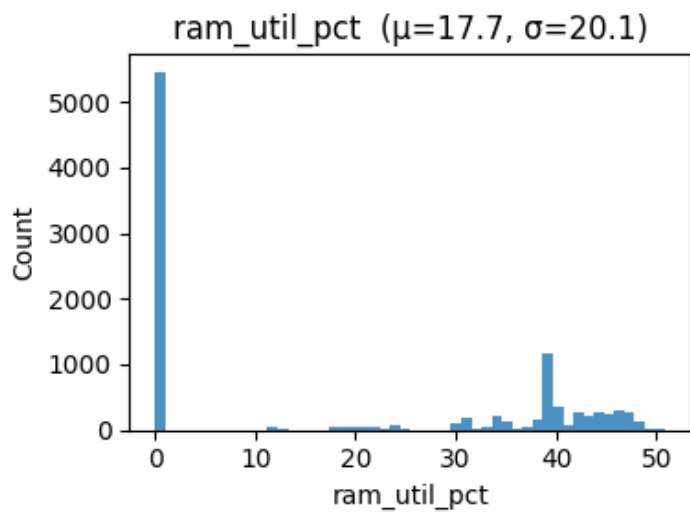


Fig 4.14 - Ram Utilisation distribution (full run)

	feat_A	feat_B	rho
75	gpu_power_pct	cpu_power_pct	+1.00
5	power_w	current_a	+0.99
54	cpu_temp_c	cpu_power_pct	+0.99
76	gpu_power_pct	cpu_temp_c	+0.98
115	ram_power_pct	cpu_temp_c	+0.97
114	ram_power_pct	cpu_power_pct	+0.93
87	gpu_temp_c	cpu_power_pct	+0.93
117	ram_power_pct	gpu_power_pct	+0.93
90	gpu_temp_c	gpu_power_pct	+0.93
88	gpu_temp_c	cpu_temp_c	+0.93
53	cpu_temp_c	cpu_util_pct	+0.91

Tab 6.1- Shap Summary Table Highly collinear pairs ($|\rho| \geq 0.9$)

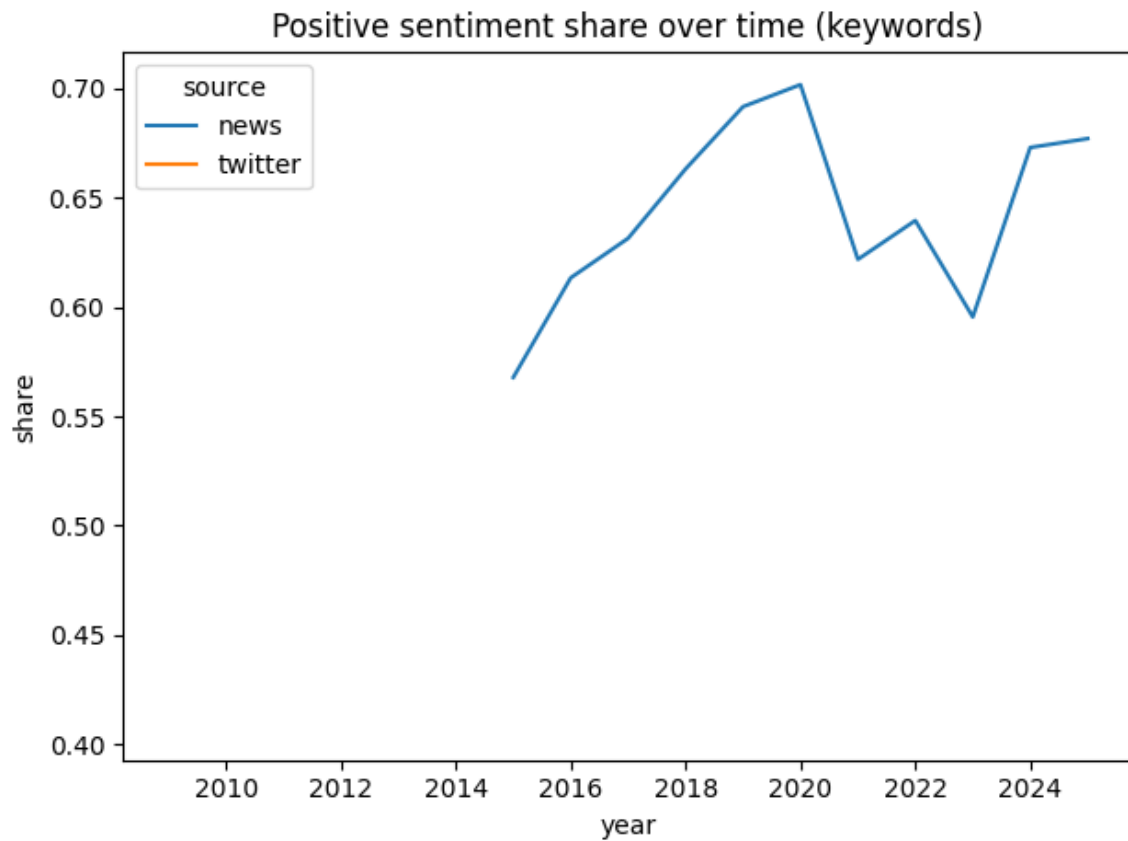


Fig. 7.2 — *Positive sentiment share over time (keywords)* – line plot for search-keyword sentiment (2015–2025).